Leading Indicators of Short-Term Inflation: A Comparative Study of Canada and the United States

STAT 443 Project Report

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1. Summary / Abstract

This project evaluates whether lagged macroeconomic indicators improve short-horizon inflation forecasts for Canada and the United States. Monthly inflation is defined as the log-difference of seasonally adjusted CPI, and forecasting performance is assessed using moving 1-step-ahead predictions over a 24-month holdout sample. Four classes of models are compared: simple benchmark rules (training-sample mean and persistence), exponential smoothing (ETS), ARIMA models capturing internal time-series dynamics, and ARIMAX models that incorporate lagged explanatory variables selected based on diagnostic evidence and economic relevance.

The results show that increased model complexity does not necessarily lead to improved forecast accuracy. For Canada, the training-sample mean provides the lowest out-of-sample RMSE, slightly outperforming ARIMA, while ETS and ARIMAX do not deliver additional gains. For the United States, ARIMA achieves the strongest performance, outperforming both benchmark rules and ETS, with ARIMAX models ranking second but failing to surpass the best univariate specification.

Although ARIMAX specifications improve in-sample fit in both countries, their weaker holdout performance suggests limited predictive value from the selected macroeconomic variables at a 1-month horizon.This finding highlights that short-run inflation is difficult to forecast using external indicators, even when diagnostic analysis suggests plausible leading relationships. Overall, the evidence favors simpler forecasting approaches: the mean forecast performs best for Canada, while a low-order ARIMA model performs best for the United States.

2. Data and Variables

2.1 Data Source

To ensure the reproducibility and reliability of the forecasts, all macroeconomic series were retrieved from official national statistical agencies and federal economic databases. The data is measured at a monthly frequency.
All domestic Canadian macroeconomic indicators were sourced directly from Statistics Canada, with the exception of the global oil benchmark.
The majority of the U.S. data, alongside the global benchmark for crude oil, was sourced via the Federal Reserve Bank of St. Louis's Federal Reserve Economic Data (FRED) database.

2.2 Target Variable

The variable of interest is monthly inflation, defined as the log difference of seasonally adjusted CPI: inflation_t = log(CPI_t) - log(CPI_{t-1}). This transformation converts the price index into a monthly inflation rate and is standard in time-series forecasting because it reduces non-stationarity in the price level

2.3 Explanatory Variables

The explanatory variables are used in lagged form so that each forecast only uses information that would have been available at the time the forecast is made.

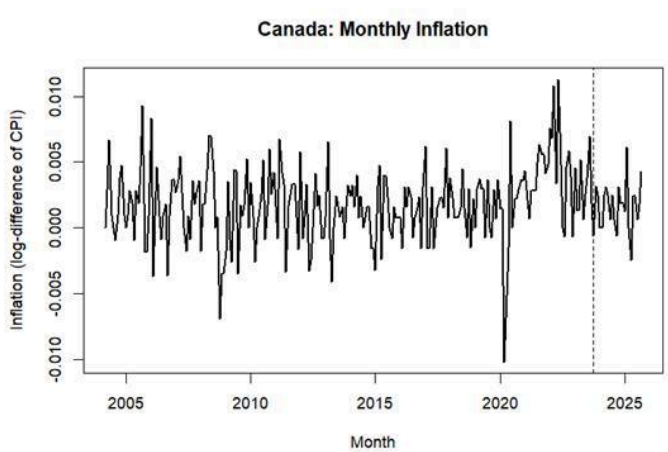
Table 1: Selection and Transformation of Explanatory Variables

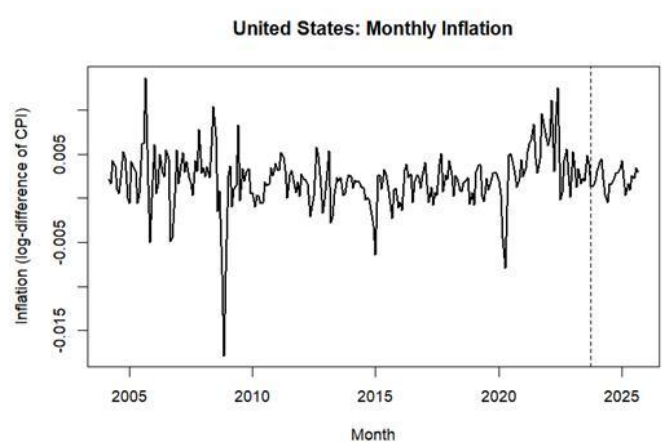
Country	Predictor Set	Measurement / Transformation
Canada	• WTI oil price • IPPI excluding energy • Real GDP • Unemployment rate • Exchange rate	Oil, IPPI, GDP, and exchange rate expressed as log changes; unemployment in levels. *All variables used at t−1*
United States	• WTI oil price • PPI excluding energy • Import price index excl. petroleum • Real GDP • U6 unemployment	Oil, PPI, and import prices as log changes; GDP is growth rate; U6 unemployment in levels. *All variables used at t−1*

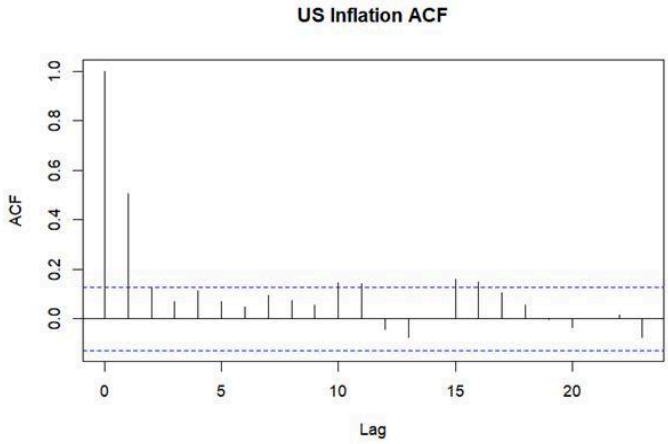
2.4 Sample Window and Forecast Split

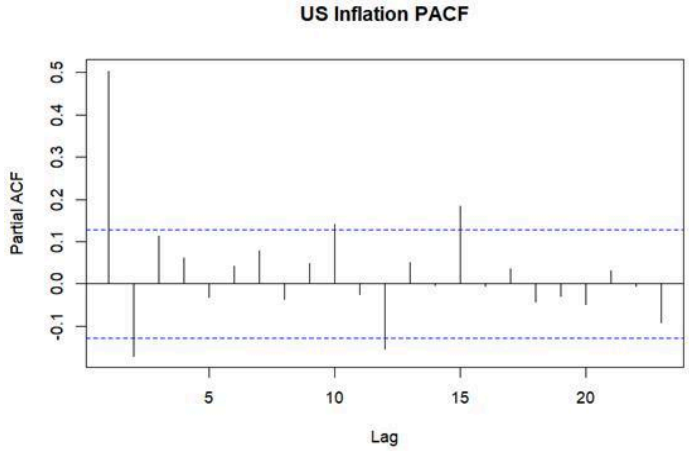
After aligning the target series and all lagged predictors, the uploaded common sample runs from March 2004 through September 2025. The final 24 months (October 2023 to September 2025) are reserved for out-of-sample evaluation, leaving March 2004 to September 2023 as the estimation window.

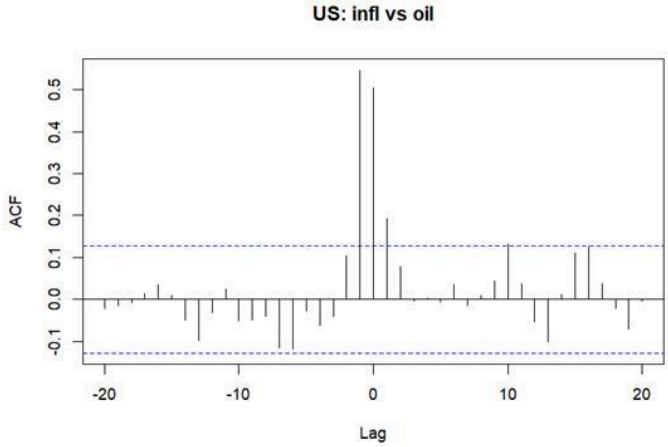
2.5 Summary Statistics

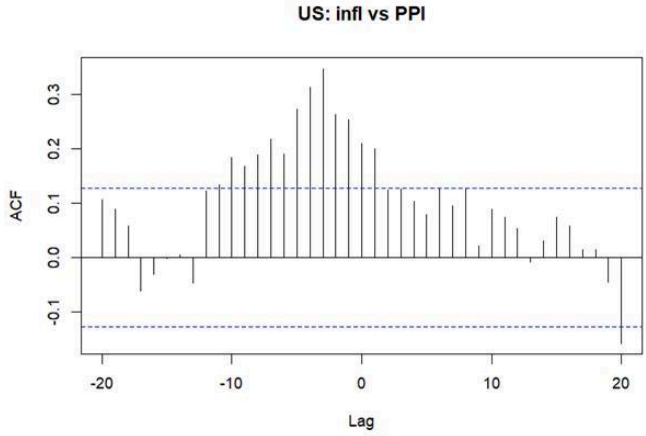





US Inflation ACF
ACF
Lag


US Inflation PACF
Partial ACF
Lag


US: infl vs oil
ACF
Lag


US: infl vs PPI
ACF
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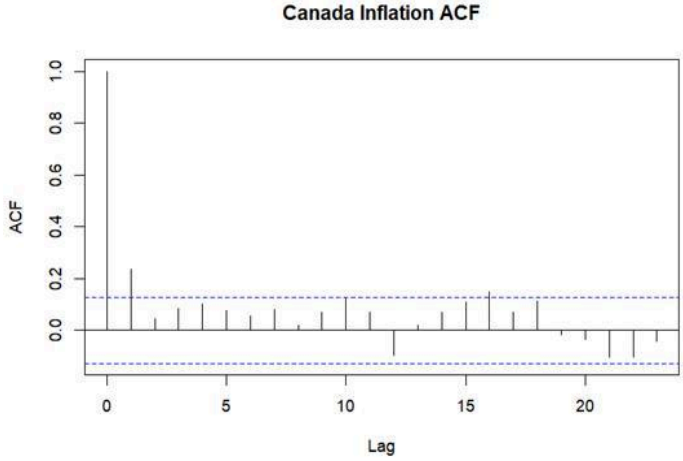
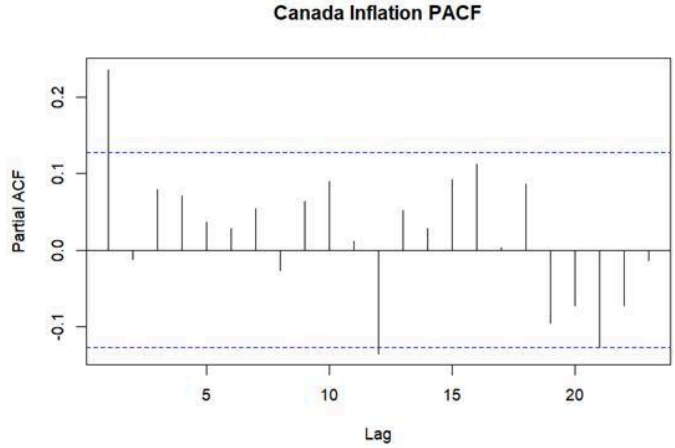

Canada Inflation ACF
ACF
Lag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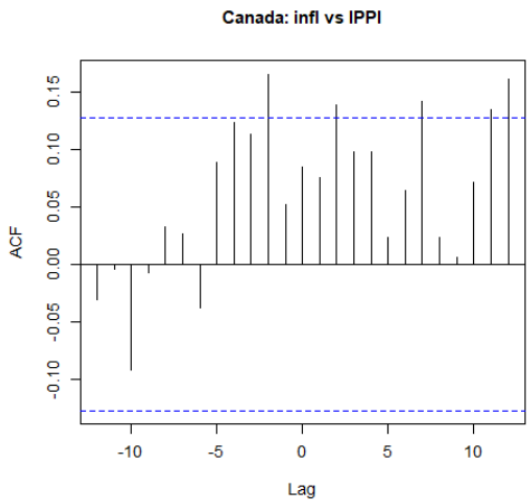


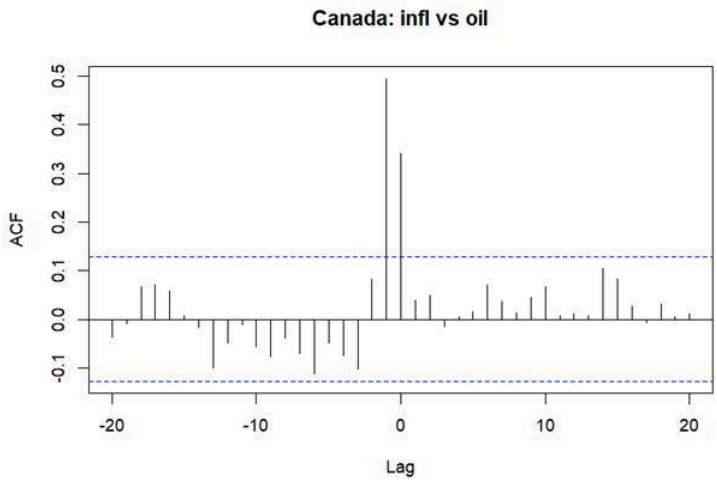


Table 2: Summary Statistics for the Transformed Variables Used in Forecasting

Canada — Summary Statistics

Variable	Unit	Mean	SD	Min	Max
Monthly inflation	log diff	0.0018	0.0028	−0.0102	0.0113
Real GDP growth (lag 1)	log diff	0.0016	0.0104	−0.1127	0.0576
Unemployment rate (lag 1)	percent	6.9019	1.2107	4.8	14.2
IPPI excl. energy (lag 1)	log diff	0.0020	0.0071	−0.0188	0.0262
WTI oil price growth (lag 1)	log diff	0.0025	0.1048	−0.5681	0.5456
Exchange rate growth (lag 1)	log diff	0.0000	0.0159	−0.0844	0.0534

United States — Summary Statistics

Variable	Unit	Mean	SD	Min	Max
Monthly inflation	log diff	0.0021	0.0032	−0.0179	0.0137

Real GDP growth (lag 1)	ann. growth rate	2.1205	7.2664	−71.98	44.68
U6 unemployment rate (lag 1)	percent	10.53	3.4112	6.6	22.9
PPI excl. energy (lag 1)	log diff	0.0022	0.0033	−0.0128	0.0137
WTI oil price growth (lag 1)	log diff	0.0025	0.1048	−0.5681	0.5456
Import price growth excl. petroleum (lag 1)	log diff	0.0009	0.0042	−0.0170	0.0146

2.6 Diagnostic Evidence from Plots

The time series plots of the log-differenced inflation for both Canada and the United States exhibit behavior consistent with stationarity. Both series fluctuate around a constant mean (near zero) and do not display long-term monotonic trends, suggesting that further regular differencing may not be strictly necessary to achieve stationarity. Both series display periods of heightened volatility, most notably the sharp, synchronous fluctuations around 2020 corresponding to the COVID-19 macroeconomic shock.

2.6.1 ACF/PACF

Examining the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots of the stationary training series, we have the following findings:

- Canada: The ACF plot for Canadian inflation shows significant positive autocorrelation at lags 1 and 2 before decaying toward zero. The corresponding PACF cuts off sharply after lag 2, with minor, marginally significant spikes at higher lags. This pattern, a decaying ACF and a truncating PACF, is highly indicative of an Autoregressive process, suggesting candidate models such as an AR(1) or AR(2).
- United States: In contrast, the US inflation diagnostics exhibit a much higher degree of persistence. The ACF decays slowly rather than dropping off immediately, and the PACF exhibits lingering significant spikes rather than a clean, sharp cut-off. Thus, it suggests a mixed Autoregressive Moving Average (ARMA) process.

Overall, This persistent, mixed autocorrelation structure in the US data, combined with the slightly cleaner AR-driven structure in the Canadian data, justifies our methodological choice to test a grid of low-order, mixed candidate models, specifically ARIMA(1,0,1), ARIMA(2,0,1), and ARIMA(1,0,2). These findings serve as the baseline for our candidate models.

2.6.2 CCF

Beyond univariate dynamics, we evaluated Cross-Correlation Function (CCF) diagnostics (see Appendix A for full diagnostic plots) between inflation and each macroeconomic covariate to identify viable leading predictors for the ARIMAX specifications. For forecasting purposes, a useful predictor must exhibit significant correlation at positive lags, indicating that the covariate's movements precede changes in inflation.

- Canada: The CCF plots reveal that while several variables show strong contemporaneous correlation (lag 0), leading signals are relatively sparse. The Industrial Product Price Index (IPPI) emerges as the strongest candidate, displaying significant positive cross-correlations extending through the first few positive lags. WTI Oil and the Exchange Rate show marginally significant spikes at lag 1, suggesting limited but potential short-term predictive utility. In contrast, Real GDP exhibits its strongest correlations at negative lags (indicating inflation leads GDP), while Unemployment lacks significant leading correlation at short horizons, disqualifying them as effective 1-step-ahead predictors.
- United States: The cross-correlations for the U.S. data demonstrate much clearer leading relationships. Both the Producer Price Index (PPI) and Import Price Index (MXP) display a broad, persistent set of significant positive spikes across multiple positive lags, indicating strong short-to-medium-term predictive power. U6 Unemployment also presents a statistically significant negative correlation at early positive lags, which aligns with theoretical macroeconomic expectations (a tighter labor market leading to higher inflation). Similar to Canada, US inflation shows weak correlation with oil prices at lag 1 and Real GDP shows a sharp contemporaneous spike but immediately drops to near-zero at lag 1, offering no leading signal.

Overall, the CCF diagnostics were used to screen plausible explanatory variables for our ARIMAX variable selection. For Canada, Industrial Product Price Index (IPPI) showed a clear positive-lag relationship, while Oil Prices displayed limited but potential short-lag association. For the US, Producer Price Index (PPI), Import Prices (MXP), and U6 Unemployment were the most plausible leading indicators at a 1-month lag (t - 1). In contrast, the strictly contemporaneous or lagging nature of Real GDP across both countries led to its exclusion from our primary forecasting specifications. However, the overall weakness and inconsistency of leading cross-correlations suggest that these predictors may have limited usefulness for very short-horizon forecasts, which is later reflected in the ARIMAX performance. These findings serve as the final baseline for our candidate model set.

3. Methodology

3.1 Forecasting Models

The project compares four classes of forecasting models.

Benchmark models:

- Mean forecast: each holdout observation is forecast using the training-sample mean inflation rate
- Persistence forecast: each forecast equals the most recently observed inflation value

Exponential smoothing: at each forecast origin, an ETS specification is estimated on the training data and produces a one-step-ahead forecast.

ARIMA: candidate non-seasonal ARIMA models

- ARIMA(1,0,1)
- ARIMA(2,0,1)
- ARIMA(1,0,2)

These candidates were chosen because the ACF and PACF diagnostics suggested stationary, low-order dynamics. Rather than relying solely on information criteria, the final ARIMA specification is selected

using holdout RMSE, which is consistent with the project's emphasis on forecast performance rather than in-sample fit. The uploaded code explicitly evaluates these three candidates for both countries.

ARIMAX: ARIMA models augmented with lagged macroeconomic predictors chosen from the diagnostic stage and screened on holdout RMSE.

For Canada, the diagnostic evidence most clearly supported industrial product prices (IPPI) as a plausible leading predictor, while oil was treated as a secondary candidate because its cross-correlation signal was concentrated mainly at contemporaneous and very short lags. Accordingly, the Canadian ARIMAX specifications considered were:

- IPPI only
- oil only
- oil and IPPI

For the United States, producer prices (PPI) showed the clearest leading relationship in the cross-correlation analysis, while oil, import prices, and U6 unemployment were treated as additional economically motivated candidates with less uniform diagnostic support. Accordingly, the U.S. ARIMAX specifications considered were:

- oil and PPI
- oil, PPI, and import prices (MXP)
- oil, PPI, import prices, and U6 unemployment.

These restricted combinations are consistent with the variables that appeared most plausible from the exploratory analysis. This restriction avoids overfitting and ensures that variable selection remains interpretable rather than data-mined. Cross-correlation diagnostics were used as a screening device to form candidate ARIMAX specifications, while holdout RMSE was used to determine whether the added regressors produced genuine forecasting gains. The code estimates the ARIMAX model recursively and generates one-step-ahead forecasts using the current information available at each step.

This recursive evaluation framework closely mimics real-time forecasting conditions, as each prediction is generated using only information available at that point in time. This design avoids look-ahead bias and provides a more realistic assessment of model performance.

3.2 Evaluation Design

The project uses moving 1-step-ahead forecasts over the holdout sample. At each origin, the model is estimated on all data available up to that month, a 1-step forecast is produced, the realized value is added to the estimation sample, and the process repeats..

Model performance is summarized with root mean square error (RMSE). Both in-sample RMSE and holdout RMSE are reported, but holdout RMSE is the main decision criterion because it measures how well the model generalizes.

4. Results

4.1 Canada

Table 3: Canada Forecast Model Performance: In-sample vs. Out-of-sample RMSE

Model	In-sample RMSE	Out-of-sample RMSE
Persistence	0.00357	0.00253
Mean	0.00288	**0.00170**
ETS	0.00285	0.00191
ARIMA(1,0,1)	**0.00280**	**0.00175**
ARIMAX(Oil + IPPI)	**0.00267**	0.00179

The mean forecast records the lowest holdout RMSE. Although ARIMAX has the best in-sample fit, its holdout error is higher than both the mean model and ARIMA, which is consistent with overfitting. This is an important result: adding macroeconomic predictors improves the within-sample fit, but that improvement does not translate into the best forecasting performance on new data

The small gap between the mean rule and ARIMA suggests that Canada inflation is persistent but not strongly structured enough for added dynamics or explanatory variables to generate economically meaningful forecasting gains at the 1-month horizon. Over the sample considered, monthly inflation appears difficult to improve upon using either richer time-series structure or the selected lagged macroeconomic indicators.

4.2 United States

Table 4: United States Forecast Model Performance: In-sample vs. Out-of-sample RMSE

Model	In-sample RMSE	Out-of-sample RMSE
Persistence	0.00327	0.00118
Mean	0.00328	0.00121
ETS	0.00315	0.00131
ARIMA(1,0,2)	**0.00278**	**0.00098**
ARIMAX(Oil + PPI + MXP + U6)	**0.00264**	**0.00109**

For the United States, the best-performing method is ARIMA. As in Canada, ARIMAX improves in-sample fit, but it does not achieve the lowest holdout RMSE. Relative to the benchmark rules, however, the U.S. results show more benefit from modeling internal time-series dynamics: ARIMA clearly outperforms mean, persistence, and ETS on the holdout sample.

Relative to Canada, the US ranking is more favorable to ARIMA-type structure and somewhat more favorable to macroeconomic regressors. The inflation series appears to contain a more exploitable short-run autocorrelation structure, but the selected external predictors still do not improve upon the best pure time-series model for one-step-ahead forecasting.

4.3 Cross-Country Comparison

A comparison of the forecasting results for Canada and the United States reveals distinct differences in the underlying dynamics of inflation and the relative effectiveness of differing models.

The diagnostic plots (section 2.5) indicated that the US inflation exhibits higher persistence as seen by the slower decaying ACF and spikes in the PACF. Despite this, the US models achieved lower holdout RMSE values compared to the Canadian models. This suggests that the US inflation is more stable and predictable using autoregressive models. In contrast, Canadian inflation seems to be more sensitive to short term volatility and thus harder to predict using univariate models.

The ranking of models shows that for the US models, the ARIMA models resulted in a better out-of-sample model than complex multivariate models such as ARIMAX models. Meanwhile, Canada models showed that simple average rules obtained results on par with ARIMA and ARIMAX models. This suggests that Canadian inflation may be less driven by immediate past inflation and contains less exploitable short-run structure at a 1-month horizon.

5. Discussion and Conclusion

5.1 Discussion

Our main finding is that increasing model complexity does not lead to better short-term inflation forecasts. Across both countries, simpler models consistently matched or outperformed more complex specifications in the holdout period.

For Canada, the mean forecast achieved the lowest RMSE, with ARIMA performing similarly. This suggests that at a 1-month horizon, inflation is largely unpredictable beyond its average level. Although ARIMAX improved in-sample fit, its weaker holdout performance indicates overfitting rather than true predictive gain.

For the United States, ARIMA performed best, indicating that short-run inflation dynamics are better captured by internal persistence than by the external predictors considered here. Compared to Canada, U.S. inflation exhibits a more stable autoregressive structure, which is consistent with the slower decay in the ACF and more persistent PACF behavior.

The cross-correlation analysis helps explain why ARIMAX underperforms. Many predictors show weak relationships or peak correlations at negative lags, suggesting that inflation may lead these variables rather than respond to them. As a result, including these predictors does not improve one-step-ahead forecasts.

Overall, the results suggest that at very short horizons, simple time-series models are difficult to beat unless predictors contain strong and consistent leading information. These findings emphasize the importance of model simplicity and robustness in short-horizon macroeconomic forecasting.

5.2 Limitations

Several limitations should be considered:

- The analysis focuses only on a 1-step-ahead horizon, where predictors may have limited impact.
- All predictors are included at lag 1. Alternative or distributed lag structures may perform better.
- The sample includes structural shocks, particularly during COVID, which may reduce model stability.

- Model comparison relies on RMSE. Other metrics such as MAE could change rankings.
- The predictor set is limited and does not fully include variables like inflation expectations.

These limitations suggest that the results apply specifically to the current design and horizon, rather than ruling out macro predictors more generally.

5.3 Conclusion

This study examined whether lagged macroeconomic indicators improve short-term inflation forecasting in Canada and the United States. Using a common dataset and recursive 1-step-ahead evaluation, the results show that simpler models perform best out of sample. The mean forecast performs best for Canada, while ARIMA performs best for the United States. Although ARIMAX improves in-sample fit, it does not consistently improve forecasting accuracy.

The key takeaway is that short-horizon inflation is difficult to predict, and model stability matters more than model complexity. In practice, simple benchmarks and low-order ARIMA models provide reliable performance, while multivariate models are only useful when predictors provide clear and stable leading signals.

6. Contributions

6.1 Team Formation

Tejasvi and Siddarth had worked on projects before and scouted Kevin and Nolan from Piazza. The group had shared interests in the project being centered around Economics since the first meeting.

6.2 Ordering of Author Names

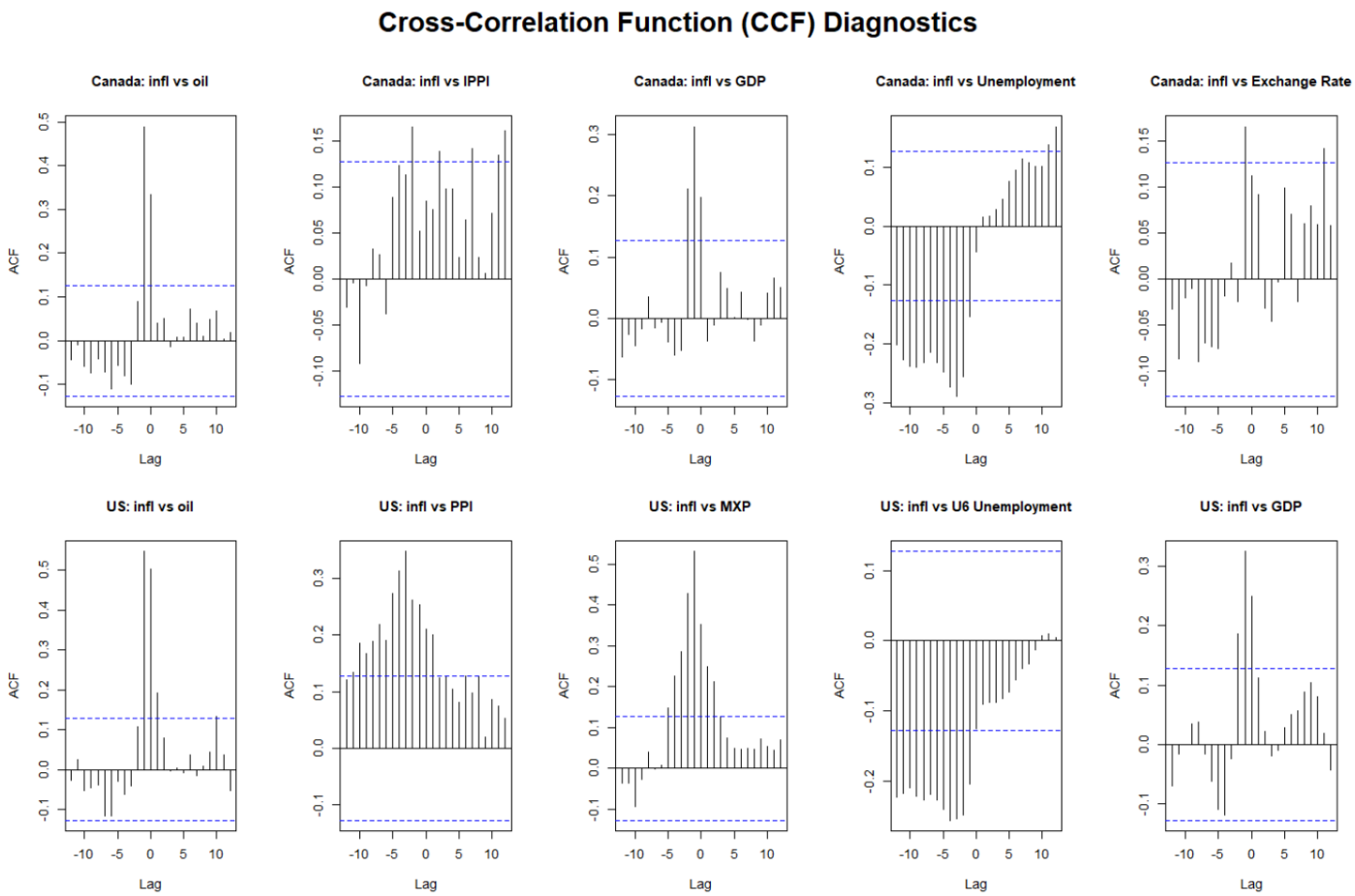
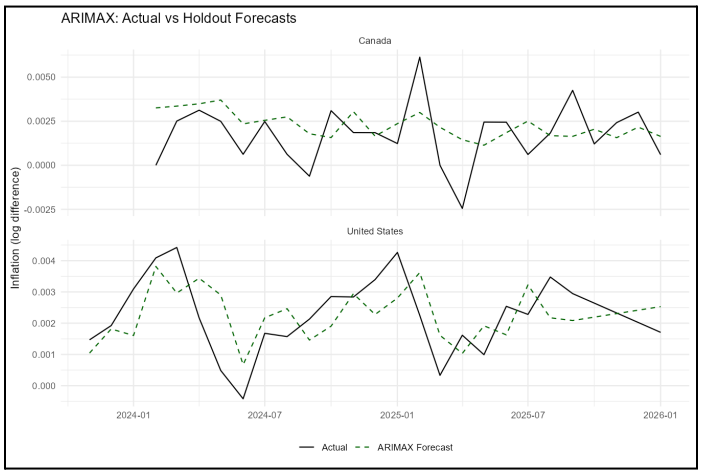
Alphabetical (Ascending Last Name)

6.3 Major Contributions

- Kevin Uli: Wrote the visual/plot diagnostic interpretations justifying model selection (Section 2.6) and the cross-country comparative analysis (Section 4.3). Additionally, refactored the project's codebase into a modular five-file R directory.
- Tejasvi Kathuria: Introduced the topic, found data sources, extracted and organized the data from sources, worked on various parts of the report from introduction to methodology and results, coding the time series models
- Siddarth Chilukuri: Worked on model implementation and analysis, including fitting ARIMA, ARIMAX, and ETS models and producing holdout forecast results (RMSE tables, predictions, and diagnostic plots). Also helped with data preparation and contributed to the report (Section 5)
- Nolan Vu: Contributed to refining the methodology section. Improved the clarity of the evaluation design by clarifying the recursive forecasting framework and its implications.

Appendix

A. CCF plot of all covariates.



B. ARIMAX: Actual vs Holdout